

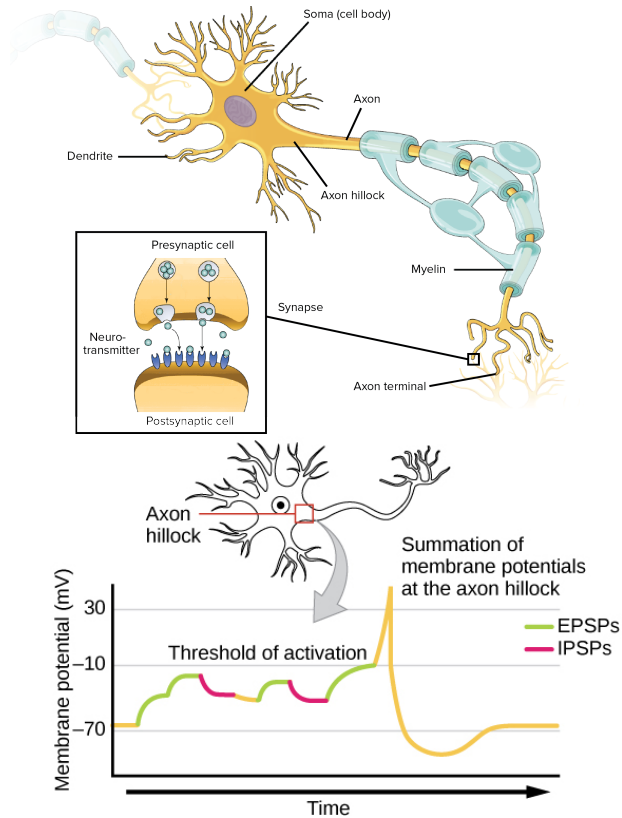
What to expect in the course

Vikas Dhiman

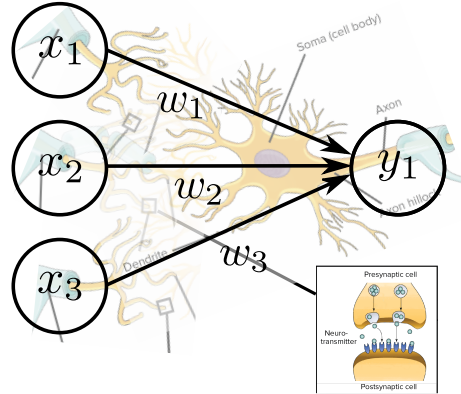
Tuesday 7th October, 2025



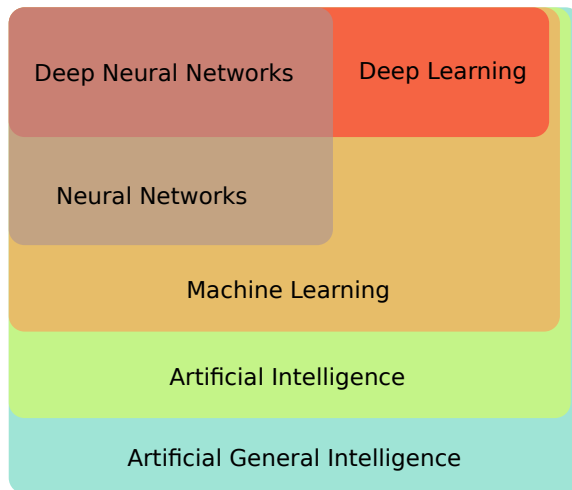
1 What is a Neural Network?



$$y_1 = \sigma(w_1x_1 + w_2x_2 + w_3x_3 + w_4)$$



2 How does it relate to other terms in the field



2.1 Comparison with other courses

- ECE 491/591: Deep Learning (Dr Yifeng Zhu)
 - We will dive deeper on backpropagation. We will implement your on Pytorch library.
 - We are more theoretical, less towards application than ECE 491.
- COS 475/575: Machine Learning (Dr Chaofan Chen)
 - There are other kinds of ML algos (SVM, Gaussian Processes, Decision trees, Random Forests, Nearest Neighbors).

- We will be focused on Neural Networks.
- COS 470/570: Introduction to Artificial Intelligence
 - Searching for solutions
 - Neural Networks are more about pattern matching.
- COS 498/598: Explainable AI (Dr Chaofan Chen)
 - Focuses on explainability of Neural Networks

3 Syllabus

Linked here: [Syllabus](#)